

CPR+ / WHITEWALKER CAPABILITY INVENTORY

Endpoint Network State Detection, Remediation, and Telemetry Framework

A truthful use-case breakdown of what CPR+ does, why it exists, and how the framework coordinates DHCP/unlock triggers, posture recovery, captive portal handling, VPN hairpin prevention, Haystack monitoring, logging, and self-update.

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1. Executive summary

CPR+ / WhiteWalker is an endpoint-side network-state framework. It is launched by Windows Task Scheduler when a device receives a DHCP address or when the user unlocks the endpoint. The main entry point, `WW_main.ps1`, waits for the network stack to settle, classifies the endpoint's real network state, and then routes to the smallest reasonable action.

Plain-English definition: CPR+ lets an endpoint answer the question, "Where am I, what network condition am I actually in, and what should I do next without making things worse?"

The framework covers more than captive portal remediation. It handles Cisco ISE posture redirect recovery, walled-garden portal detection, VPN state handling, on-prem VPN hairpin prevention, trusted SSID tie-breaking, Haystack event monitoring, syslog queueing, local diagnostics, state tracking, and gated self-update.

2

Primary launch triggers: DHCP/address change and unlock

40+

Operational behaviors and remediation paths documented here

200+

Reported Pester test cases across major components

Architecture flow

Primary use cases

Capability matrix

Haystack and telemetry

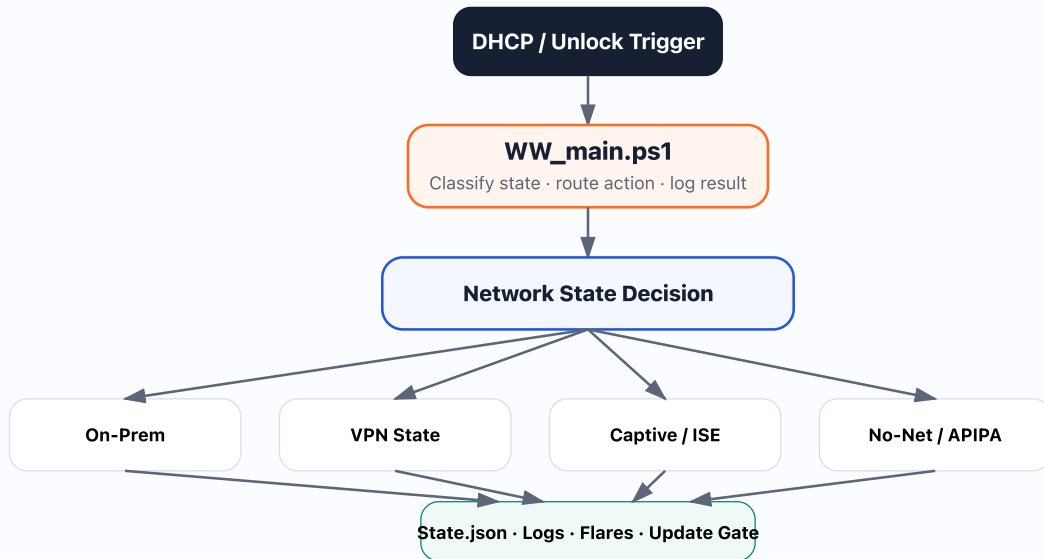
Safety controls

Summary list

Decision trees

2. Framework architecture

The root of the framework is `WW_main.ps1`. Supporting scripts own specific actions, so the main script can stay focused on classification, decision routing, state updates, and logging.



Trigger



Detect / classify



Remediate, log, or exit cleanly

3. Primary use cases covered

Network access recovery

- ISE employee posture redirect detection
- Posture rescan trigger and compliance verification
- Captive portal browser launch
- Guest portal and walled-garden handling

- APIPA / no-network safe exit

VPN behavior control

- User tunnel and management tunnel detection
- VPN stuck/intermediate state handling
- SSID-change checks while VPN is active
- On-prem VPN hairpin prevention
- Hosts-file blackhole cleanup on non-on-prem states

Operations telemetry

- State file maintained for downstream scripts
- FlareGun event routing
- Haystack needle monitoring
- Syslog/Splunk forwarding with offline queue
- Diagnostic and tail helper scripts

4. Capability matrix

Area	What CPR+ does	Operational value
DHCP / unlock execution	Runs automatically after address changes and unlock events, then waits briefly for the network stack to settle.	Catches the moment where most endpoint network failures show up without requiring the user to launch a tool.
State classification	Classifies on-prem, off-prem, user tunnel, management tunnel, captive portal, walled garden,	Prevents one-size-fits-all remediation. The endpoint

Area	What CPR+ does	Operational value
	untrusted SSID, and transient no-network states.	acts based on what is actually happening.
ISE posture recovery	Detects posture redirects, triggers Cisco posture rescan, watches for compliance, and verifies post-compliance access.	Reduces tickets where the endpoint is connected but trapped behind posture ACLs.
Captive portal remediation	Routes captive portal handling into the user context so browser and toast workflows happen where the user can interact.	Fixes hotel, guest, and venue networks without SYSTEM-context browser weirdness.
Walled-garden detection	Identifies networks where the gateway is reachable but DNS or HTTPS validation is blocked until portal acceptance.	Avoids mislabeling hotel-style captive portals as simple internet failure.
VPN hairpin prevention	When safely confirmed on-prem, sinkholes VPN headend names to localhost using a managed hosts-file block.	Prevents unnecessary VPN connection attempts from inside the corporate network.
Trusted SSID tie-breaker	Uses hardcoded and synced trusted SSID lists to avoid false on-prem detection from stale VPN routes or misleading DC reachability.	Prevents CPR+ from disconnecting or blocking VPN while the user is really off-prem.
Haystack monitoring	Watches for configured Windows Event Log needles and writes structured hits.	Adds lightweight endpoint event detection without a heavy always-polling agent loop.

Area	What CPR+ does	Operational value
Syslog queueing	Sends structured events to syslog/Splunk when reachable; queues locally when offline and drains later.	Keeps telemetry useful without breaking endpoint workflows when the collector is unreachable.
Self-update	Uses DEV/PROD flag files and network-state gating to pull updates from NETLOGON with cooldown controls.	Supports controlled repair and rollout without touching every endpoint manually.
Logging and diagnostics	Writes decision breadcrumbs, state changes, warnings, exit reasons, and rotating logs.	Gives IT teams evidence instead of guesses.

5. Detailed use case inventory

DHCP / unlock network state evaluation

CPR+ detects the active network context after DHCP and unlock events. It checks preflight behavior, redirect signals, adapter state, VPN state, SSID trust, and gateway/DNS conditions before deciding what to do.

on_prem

user_tun

mgmt_tun

off_prem_no_vpn

captive_portal

no_net_transient

ISE posture rescan automation

When posture redirect behavior is detected, CPR+ routes into the posture workflow, triggers rescan, checks service availability, monitors compliance, and verifies that network access actually recovered.

Captive portal and guest Wi-Fi

Portal handling is intentionally moved into a user-context runner. That keeps the main script from launching browsers as SYSTEM and gives the user a usable auth window.

Walled-garden networks

CPR+ distinguishes DNS-blocking portal conditions from general network failure. Gateway reachable plus DNS failure can route to the portal browser instead of dumping into a dead-end no-network state.

VPN blackhole framework

When the device is confirmed on-prem, CPR+ can apply a controlled hosts-file sinkhole for configured VPN headends. Removal is deliberately broad and safety-biased on startup, shutdown, off-prem, VPN, captive, and no-network states.

Wake and SSID-change recovery

After wake or Wi-Fi movement, VPN can appear connected while blocking the local auth path. CPR+ checks SSID changes, gives VPN a grace window, and can disconnect only when it appears stuck or harmful.

6. Haystack, syslog, and telemetry

Haystack is the event-monitoring side of CPR+. It watches for configured “needle” patterns in Windows Event Logs and writes structured hits. Those hits can be sent to syslog/Splunk when the network state says the collector should be reachable.

Important distinction: Haystack is not the root decision engine. `WW_main.ps1` detects current network state and enables network-aware decisions. Haystack consumes that state so telemetry can be sent, cached, or deferred safely.

Haystack behavior	Why it matters
Merges local and shared needle files	Allows local testing before promoting a shared detection pattern.
Task Scheduler event wakeup	Avoids a constant polling loop.
Structured <code>NEEDLE_HIT</code> output	Makes downstream parsing sane.
Offline syslog cache	Prevents data loss when the endpoint is off-net or collector is unreachable.
FIFO drain with retention limits	Keeps queue behavior bounded and predictable.

7. Safety and “do no harm” controls

CPR+ is built around conservative cleanup and cooldown behavior. The general pattern is: detect carefully, act narrowly, log clearly, and remove risky state aggressively when conditions change.

- Blackhole removal on startup
- Blackhole removal on all non-on-prem states
- Shutdown task to clear VPN sinkholes before reboot
- Trusted SSID tiebreaker before on-prem actions
- Cooldowns for flares, rescans, updates, and syncs
- Local cache fallback when NETLOGON is unreachable
- Version and hash checks before config replacement
- Size-based log rotation
- WhatIf, debug, and trace modes
- APIPA/no-network early exits
- Portal and browser handling isolated into user context

- Self-update gated by current network state

8. Test and validation coverage

The repo documentation reports roughly 200+ Pester tests across the major scripts, with specific focus on the main decision flow, flare handlers, captive portal runner, diagnostics, browser handling, state management, and smart wait behavior.

~40

WW_main.ps1 tests reported

~70

Captive portal runner tests reported

Tested area	Examples
Main logic	Wake detection, flare behavior, state management, preflight classification, VPN status parsing.
Captive portal runner	Browser launch, PID tracking, smart wait polling, validation handling, Cisco browser cleanup.
Blackhole behavior	Add/remove paths, disabled behavior, startup safety removal.
Diagnostics	VPN status parsing and diagnostic collection paths.

9. Summary list: what CPR+ does

1. Runs on DHCP/address change and unlock.
2. Classifies current endpoint network state.

3. Detects clean network access through preflight behavior.
4. Detects ISE posture redirect conditions.
5. Triggers Cisco posture rescan.
6. Verifies compliance and post-compliance access.
7. Detects standard captive portals.
8. Detects ISE guest portals.
9. Detects DNS-blocking walled gardens.
10. Launches portal remediation in user context.
11. Tracks portal remediation state.
12. Handles portal failure and timeout paths.
13. Detects user VPN tunnel state.
14. Detects management VPN tunnel state.
15. Recognizes VPN intermediate/stuck states.
16. Handles SSID changes while VPN is active.
17. Disconnects VPN only when it appears to block local auth.
18. Prevents on-prem VPN hairpinning.
19. Applies managed hosts-file sinkholes for VPN headends.
20. Removes sinkholes on startup, shutdown, and non-on-prem states.
21. Supports AD allow-group exemptions for blackhole behavior.
22. Syncs blackhole configuration from NETLOGON.
23. Uses trusted SSID tie-breaking to avoid false on-prem actions.
24. Syncs trusted SSID config with local fallback.
25. Detects APIPA and no-usable-network conditions.
26. Waits for corporate Wi-Fi address assignment when appropriate.
27. Logs MAC randomization state and warnings.
28. Cleans up interfering Cisco browser/helper processes.
29. Fires structured FlareGun events.
30. Maintains `state.json` for downstream scripts.
31. Runs Haystack event monitoring when enabled.

- 32. Matches Event Log needles from config.
- 33. Writes structured `NEEDLE_HIT` records.
- 34. Forwards hits to syslog/Splunk.
- 35. Caches syslog hits offline and drains later.
- 36. Supports DEV/PROD self-update flags.
- 37. Gates updates by network state and share reachability.
- 38. Uses cooldowns to prevent loops and share thrash.
- 39. Rotates logs by size.
- 40. Provides useful local evidence for IT troubleshooting.

10. Expandable decision trees

The trees below are collapsed by default so the main report stays readable. They are intended for reviewers, support engineers, and the brave few who actually want to see how the sausage gets made.

Reading note: These are architecture-level decision trees, not a line-by-line code dump. They preserve the intent of the CPR+ flows while avoiding every internal helper call and guard clause.

Main state engine

ISE posture

Captive portal

VPN blackhole

Haystack

Syslog drain

Self-update

FlareGun

Safety cleanup

Trusted SSID

WW_main.ps1 — primary state classification

Triggered by DHCP/address change or user unlock. The script waits for the network stack, collects context, classifies state, writes breadcrumbs, then exits.

DHCP / unlock trigger

Wait for usable network context

Run preflight and local checks

Classify endpoint state

Clean access

- Gateway/preflight returns expected response.
- Use SSID and VPN checks to avoid false on-prem classification.
- Write state and route to on-prem/off-prem/VPN handling.

Redirect or blocked access

- Inspect redirect behavior.
- Separate ISE posture, ISE guest, generic captive portal, and walled garden cases.

- Fire the correct remediation event.

No usable path

- Check APIPA, missing gateway, DNS failure, and VPN intermediate states.
- Use wait/grace paths before declaring failure.
- Log clear exit reason.

ISE posture remediation decision tree

Handles employee posture redirect conditions without asking the user to reboot or blindly reconnect.

ISE employee redirect detected

Check Cisco posture service

Trigger rescan

Verify ACL lift / access restored

Service available

- Kick posture rescan.
- Monitor compliance result.
- Re-probe network access after compliance.

Service unavailable

- Log posture service failure.
- Fire service-unavailable flare.
- Avoid looping rescans.

Compliance fails

- Record failure state.
- Leave useful evidence for desktop/network teams.
- Do not mask the failure with unrelated remediation.

Captive portal and walled-garden decision tree

Separates normal portal redirects from DNS-blocking hotel/venue networks and ISE guest portals.

Preflight redirect / blocked validation

Classify portal type

Launch user-context runner

Validate cleared or timeout cleanly

Standard captive portal

- Detect HTTP redirect.
- Open browser in user context.
- Track portal state and completion.

ISE guest portal

- Recognize guest redirect signature.
- Route to portal runner rather than posture rescan.
- Log as guest/captive path.

Walled garden

- Gateway reachable but DNS/preflight blocked.
- Open gateway/portal path directly.
- Avoid false generic no-network classification.

VPN hairpin / blackhole decision tree

The blackhole framework is intentionally safety-biased: only add sinkholes when on-prem is strongly supported; remove them broadly everywhere else.

Current state evaluated

Confirm on-prem and not exempt

Apply managed hosts-file sinkhole

Remove on any unsafe state

Add path

- Device confirmed on trusted on-prem network.
- User/device not in allow/exempt group.
- VPN headend names mapped to localhost.

Remove path

- Startup or shutdown.
- Off-prem, VPN, captive portal, APIPA, or no-network.
- Unknown state where leaving sinkhole could strand the user.

Config path

- Sync host list and allow groups from NETLOGON.

- Use version/hash checks and cooldowns.
- Fall back to local cache.

Trusted SSID tie-breaker decision tree

Prevents stale routes or misleading DC reachability from making an off-prem Wi-Fi look like corporate LAN.

Potential on-prem signal

Check adapter and SSID context

Trust / distrust decision

Allow or block on-prem actions

Trusted wired/corp Wi-Fi

- Proceed with on-prem classification.
- Allow blackhole and on-prem state actions.

Untrusted Wi-Fi

- Treat DC reachability cautiously.
- Avoid killing or blocking VPN.
- Classify as off-prem/untrusted where appropriate.

Config handling

- Use built-in trusted values plus synced list.
- Cooldown sync attempts.
- Use cache if share is unavailable.

Haystack needle handling decision tree

Haystack is the event-monitoring subsystem. It wakes on event triggers, checks configured needles, logs structured hits, and hands forwarding to the syslog path.

Event Log trigger

Load merged needle set

Match within lookback window

Write NEEDLE_HIT / queue forward

Needle match

- Normalize the event context.

- Write structured local log output.
- Send or queue telemetry.

No match

- Exit without noisy action.
- No continuous polling loop.

Needle config

- Merge local test needles with shared config.
- Support safe local testing before promotion.
- Use bounded lookback to avoid missing near-trigger events.

Syslog / Splunk cache-and-drain decision tree

Telemetry should be useful when reachable and harmless when not. The queue keeps endpoint actions from depending on collector availability.

Telemetry event ready

Check current network state

Send now or cache

Drain FIFO when reachable

Collector reachable

- Send structured syslog message.
- Drain cached queue in order.

Collector unreachable

- Write to local queue.
- Do not block remediation.
- Respect retention/max-entry bounds.

State-aware drain

- Drain on on-prem, user tunnel, or management tunnel states.
- Skip drain on captive/no-network states.

Self-update decision tree

The updater is gated so repair and rollout do not become a share-thrashing stampede.

DEV / PROD flag detected

Check network/share reachability

Apply update if eligible

Record result and cooldown

Eligible

- Network state allows share access.
- Cooldown permits update.
- Pull package/config from trusted share path.

Not eligible

- Off-net, captive, or share unavailable.

- Leave flag/result evidence.
- Try later when state changes.

Safety

- Separate DEV and PROD paths.
- Avoid loops with cooldown markers.
- Log success/failure plainly.

FlareGun event-routing decision tree

FlareGun lets CPR+ separate detection from action, especially when an action must run in the user context instead of SYSTEM.

State/action event raised

Determine SYSTEM vs USER context

Write event / trigger task

Handler performs bounded action

SYSTEM-context event

- Telemetry, state write, update gate, or service-level action.
- Runs without user UI assumptions.

USER-context event

- Browser launch, toast, or portal interaction.
- Routed through Event Log / Task Scheduler.

Noise control

- Cooldown repeated events.
- Record event source and reason.
- Avoid duplicate user prompts.

Startup, shutdown, and cleanup safety tree

This is the “don’t leave the machine booby-trapped” path. Anything that can strand a user gets a broad cleanup route.

Startup / shutdown / unsafe state

Identify persisted risky state

Remove sinkholes / stale flags

Log cleanup and exit

Hosts-file safety

- Remove VPN sinkhole block on startup/shutdown.
- Remove on non-on-prem states.

Flag safety

- Clean stale portal/update markers when appropriate.
- Respect cooldowns to prevent thrash.

Logging safety

- Rotate logs by size.
- Leave clear breadcrumbs for support.

- Prefer clean exit over mystery behavior.

11. Bottom line

Best concise description: CPR+ is an endpoint network-state remediation and telemetry framework. It reduces support ambiguity by letting the endpoint determine its real access condition, take bounded corrective action, and leave behind readable evidence for IT.

Its strongest use cases are corporate Wi-Fi / ISE posture recovery, captive portal handling, VPN hairpin prevention, wake/network-change cleanup, endpoint-side event detection through Haystack, and safer field rollout through config sync plus self-update.

Generated as a CPR+ capability inventory draft. Logo loaded from the public WhiteWalker repository docs directory. This document intentionally avoids sales claims and sticks to observed/reported framework behavior.